



Barker
College

2021

YEAR 12

ASSESSMENT TASK 3

Mathematics Advanced

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Student Number

Staff Involved:

• AYG* • LZM • LAK • DXC • PJR

• RJW • PDJ • DZP • ARP

Wednesday 2nd June, 8:30am

135 copies

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General Instructions

- Working time – 55 minutes
- Write using blue or black pen
- Write your Barker Student number in the spaces indicated
- Write your answers in the spaces provided
- Board approved calculators may be used
- The HSC Reference Sheet is provided.
- Attempt all questions
- ALL necessary working must be shown
- Marks may not be awarded for careless or badly arranged work
- Diagrams are NOT to scale

Topics Covered

- Exponential and Logarithmic Functions
- Trigonometric Functions

Section	Marks
Question 1	/ 12
Question 2	/ 15
Question 3	/ 12
Question 4	/ 12
Total Marks:	/ 51

Question 1 (12 marks)

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Student number

(a) Differentiate with respect to x

(α) $y = \ln(7x - 2)$ 1

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(β) $y = (e^x + x)^3$ 2

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(γ) $y = \frac{\cos x}{x^2}$ 2

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(b) Find the following indefinite integrals:

(α) $\int \cos 5x \, dx$ 1

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(β) $\int \frac{1-2x^5}{x} \, dx$ 2

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(c) Evaluate, writing your answer in simplest form.

(α) $\int_1^{e^3} \frac{5}{x} \, dx$ 2

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(β) $\int_0^{\frac{\pi}{8}} \sec^2 2x \, dx$ 2

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Question 2 (15 marks)

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Student number

(a) Find the equation of the normal to the curve $y = \ln(x - 1) + 3$ at the point (2,3) on the curve. 3

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(b) Differentiate $y = \log_{10}(\sin x)$ 2

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(c) The function $f(x)$ has derivative $f'(x) = 8 \sin(4x)$ and $f\left(\frac{\pi}{4}\right) = 1$. 3

Find $f(x)$.

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(d) (i) Find $\frac{d}{dx}(x^2 \ln x)$ 2

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(ii) Hence, or otherwise find $\int 3x(1 + \ln x^2) dx$ 2

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(e) Evaluate $\int_0^1 \frac{x-1}{x^2-2x+4} dx$ 3

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Student number

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- A graph of the function $y = \cos(x)$ is shown on a Cartesian coordinate system. The x-axis is labeled with values $0, \pi/2, \pi, 3\pi/2, 2\pi, 5\pi/2, 3\pi$. The y-axis is labeled with y . The curve starts at point A on the y-axis at $(0, 1)$, crosses the x-axis at $(\pi/2, 0)$, reaches a minimum at $(\pi, -1)$, crosses the x-axis again at $(3\pi/2, 0)$, and ends at $(2\pi, 1)$. The area between the curve and the x-axis is shaded. The region above the x-axis (from $x = 0$ to $x = \pi$) is labeled A.

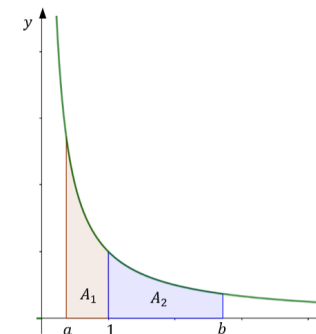
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- [illegible]

- 3

Find the values of a and b .

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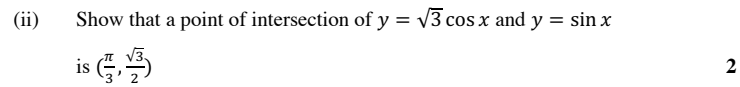
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END OF QUESTION 3

Student number

(i) Sketch the curve $y = \sqrt{3}\cos x$ below on the same axes. 1



(iii) Hence, find the simplest exact area bounded by the curve $y = \sqrt{3} \cos x$, $y = \sin x$ and the x -axis between ordinates $x = 0$ and $x = \frac{\pi}{2}$. **3**

A Cartesian coordinate system showing the graphs of $y = e^x$ (green curve) and $y = \ln x$ (red curve). The green curve is the reflection of the red curve across the line $y = x$. The area between the two curves from $x = 0$ to $x = 1$ is shaded in light blue. The intersection point P is marked at $(1, 1)$. The x-axis is labeled x and the y-axis is labeled y .

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END OF ASSESSMENT

2021 Year 12 Mathematics
Advanced Assessment Task 3
Student Solutions

Question 1.

a) α $y = \ln(7x-2)$

$$\frac{dy}{dx} = \frac{7}{7x-2}$$

β $y = (e^x + x)^3$

$$\frac{dy}{dx} = 3(e^x + x)^2 \times (e^x + 1)$$

$$= 3(e^x + x)^2 (e^x + 1)$$

γ $y = \frac{\cos x}{x^2}$ $u = \cos x$ $u' = -\sin x$

$$\therefore \frac{dy}{dx} = \frac{-\sin x \cdot x^2 - 2x \cdot \cos x}{x^4}$$

$$v = x^2$$

$$v' = 2x$$

$$= \frac{-x(x \sin x + 2 \cos x)}{x^4}$$

$$= \frac{-(x \sin x + 2 \cos x)}{x^3}$$

b) α $\int \cos 5x \, dx$

$$= \frac{1}{5} \sin 5x + C$$

b) β $\int \frac{1-2x^5}{x} \, dx$

$$= \int \frac{1}{x} - 2x^4 \, dx$$

$$= \ln x - \frac{2x^5}{5} + C$$

c) α $\int_1^{e^3} \frac{5}{x} \, dx$

$$= [5 \ln x]_1^{e^3}$$

$$= 5 \ln e^3 - 5 \ln 1$$

$$= 5 \times 3 = 15$$

β $\int_0^{\frac{\pi}{8}} \sec^2 2x \, dx$

$$= \frac{1}{2} [\tan 2x]_0^{\frac{\pi}{8}}$$

$$= \frac{1}{2} \left(\tan \frac{\pi}{4} - \tan 0 \right)$$

$$= \frac{1}{2} (1 - 0) = \frac{1}{2}$$

Question 2.

a) Find the equation of the normal
to $y = \ln(x-1) = 3$ at $(2, 3)$

$$\frac{dy}{dx} = \frac{1}{x-1}$$

$$\therefore \text{gradient of tangent} = \frac{1}{2-1} = 1$$

$$\therefore \text{gradient of normal} = -1$$

$$\therefore y - 3 = -1(x - 2)$$

$$y - 3 = -x + 2$$

$$\boxed{y = -x + 5}$$

b) $y = \log_{10}(\sin x)$

$$\frac{dy}{dx} = \frac{1}{\ln 10 \times \sin x} \times \cos x$$

$$= \frac{\cos x}{\ln 10 \times \sin x} = \frac{\cot x}{\ln 10}$$

c) $f'(x) = 8 \sin(4x)$ & $f\left(\frac{\pi}{4}\right) = 1$

$$\therefore f(x) = \int f'(x) \, dx$$

$$= \int 8 \sin(4x) \, dx$$

$$= -2 \cos 4x + C$$

$$f\left(\frac{\pi}{4}\right) = 1 \Rightarrow 1 = -2 \cos \pi + C$$

$$\therefore 1 = 2 + C$$

$$C = -1$$

$$\therefore f(x) = -2 \cos(4x) - 1.$$

d) i) $\frac{d}{dx} (x^2 \ln x)$

$$= 2x \ln x + x^2 \times \frac{1}{x}$$

$$= 2x \ln x + x$$

$$= x(2 \ln x + 1)$$

ii) $\int 3x(1 + \ln x^2) \, dx$

$$= 3 \int x(1 + 2 \ln x) \, dx$$

$$= 3x^2 \ln x + C$$

e) $\int_0^1 \frac{x-1}{x^2-2x+4} \, dx$

$$= \frac{1}{2} \int_0^1 \frac{2x-2}{x^2-2x+4} \, dx$$

$$= \frac{1}{2} \left[\ln |x^2 - 2x + 4| \right]_0^1$$

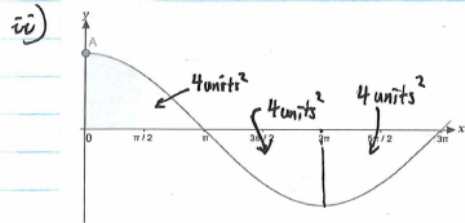
$$= \frac{1}{2} (\ln |1 - 2 + 4| - \ln 4)$$

$$= \frac{1}{2} (\ln 3 - \ln 4)$$

$$= \frac{1}{2} \ln \left(\frac{3}{4} \right)$$

Question 3.

a) i) $A = (0, 2)$



$$\therefore \int_0^{3\pi} 2 \cos \frac{x}{2} dx = 4 - 4 - 4 = -4$$

b) $\int 2^x \ln 6 dx$

$$= \frac{\ln 6 \times 2^x}{\ln 2} + C$$

$$= \frac{4 \ln 2 \times 2^x}{\ln 2} + C$$

$$= 4 \times 2^x + C$$

$$= 2^2 \times 2^x + C = 2^{x+2} + C$$

c) $\int_a^1 \frac{1}{x} dx = 1$

$$\therefore [\ln x]_a^1 = 1$$

$$\ln 1 - \ln a = 1$$

$$-\ln a = 1$$

$$\ln a = -1$$

$$\therefore a = e^{-1}$$

$$\int_1^b \frac{1}{x} dx = 1$$

$$[\ln x]_1^b = 1$$

$$\ln b - \ln 1 = 1$$

$$\ln b = 1$$

$$b = e^1 = e$$

d) i) $\log_x 2 = \frac{1}{\log_2 x}$

$$\text{LHS} = \log_x 2$$

$$= \frac{\log_2 2}{\log_2 x} = \frac{1}{\log_2 x} = \text{RHS}$$

ii) $\log_2 x = 4 \log_x 2$

$$\therefore \log_2 x = \frac{4}{\log_2 x}$$

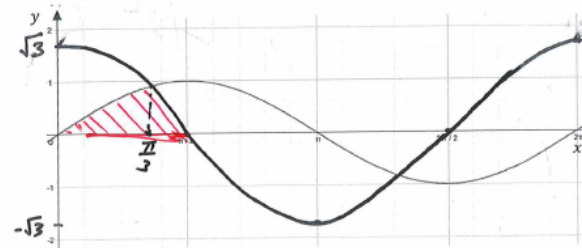
$$(\log_2 x)^2 = 4$$

$$\therefore \log_2 x = \pm 2$$

$$\therefore x = 2^2 = 4 \text{ or } x = 2^{-2} = \frac{1}{4}$$

Question 4.

a) i)



ii) $\sqrt{3} \cos x = \sin x$ $\left| \begin{array}{l} y = \sin(\frac{\pi}{3}) \\ \frac{\sin x}{\cos x} = \sqrt{3} \\ \tan x = \sqrt{3} \end{array} \right. \left| \begin{array}{l} = \frac{\sqrt{3}}{2} \\ \therefore (\frac{\pi}{3}, \frac{\sqrt{3}}{2}) \end{array} \right.$
 $\therefore x = \frac{\pi}{3}$ is a point of intersection.

iii) $\text{Area} = \int_0^{\frac{\pi}{3}} \sin x dx + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sqrt{3} \cos x dx$

$$= [-\cos x]_0^{\frac{\pi}{3}} + \sqrt{3} [\sin x]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= \left(-\frac{1}{2} + 1\right) + \sqrt{3} \left(1 - \frac{\sqrt{3}}{2}\right)$$

$$= \frac{1}{2} + \sqrt{3} - \frac{3}{2} = (\sqrt{3} - 1) \text{ units}^2$$

b)

i) $4e^{-x} = e^x - 3$

$$\frac{4}{e^x} = e^x - 3$$

$$4 = e^{2x} - 3e^x$$

$$e^{2x} - 3e^x - 4 = 0$$

$$\therefore (e^x + 1)(e^x - 4) = 0$$

$$\therefore e^x = -1 \text{ or } e^x = 4$$

No solution. $x = \ln 4$

$\therefore$ Only one point of intersection at $x = \ln 4$.

ii) $\text{Area} = \int_0^{\ln 4} 4e^{-x} - (e^x - 3) dx$

$$= \int_0^{\ln 4} 4e^{-x} - e^x + 3 dx$$

$$= [-4e^{-x} - e^x + 3x]_0^{\ln 4}$$

$$= (-4e^{-\ln 4} - e^{\ln 4} + 3 \ln 4) - (-4e^0 - e^0 + 3 \times 0)$$

$$= -4 \times \frac{1}{4} - 4 + 3 \ln 4 + 5$$

$$= -1 - 4 + 3 \ln 4 + 5$$

$$= 3 \ln 4 \text{ units}^2$$

$$= 6 \ln 2 \text{ units}^2$$